

# TFPT — Appendix H: the horizon unit system

One seam constant  $c_3 = \frac{1}{8\pi}$  as the universal horizon thermal code  
(Hawking, de Sitter, Unruh, Page, scrambling — *standard physics in seam units, not new gravity*)

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## What this document is about

A change of bookkeeping, not new gravitational physics: if gravity is the geometry-channel readout of the seam, then *all* horizons read the same boundary constant  $c_3 = \frac{1}{8\pi}$ . This note collects the readouts — Hawking, de Sitter and Unruh temperature, black-hole thermodynamics, the Page time, scrambling, the Nariai bound,  $v_{\text{GW}} = c$  and cosmic birefringence — in seam units, with two genuine compiler fingerprints (1920 =  $|W(D_5)|$ ,  $|\mu_4| = 4$ ).

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## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}$ ,  $G_{\text{net}}$ ,  $F_{\text{transfer}}$ ) as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the

universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

## You are reading companion H (Appendix H — Horizon Readouts).

### TFPT status at a glance — read this card the same way in every document

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed (**[E]**); the everything-open residual reduces to *three named interface problems*:

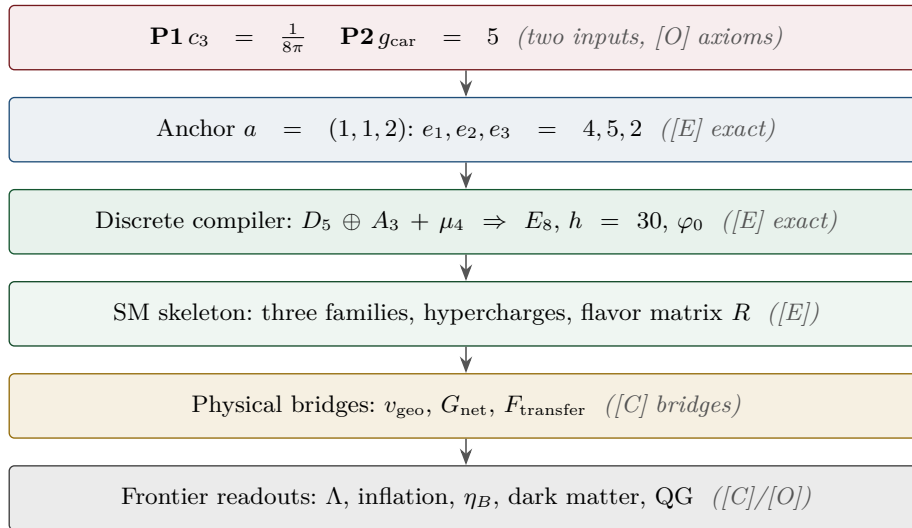
| Interface             | Question                                                                                        | Status                  |
|-----------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive <b>[O]</b>    |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | <b>[E]</b> / <b>[C]</b> |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | <b>[C]</b>              |

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ . Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**[E]** exact / machine-proven

**[C]** conditional

**[O]** open / axiom

**[X]** kill test


**Scope and honest typing.** If gravity is the geometry-channel readout of the seam (companion note “Seam response grammar”), then horizons are not separate miracles — they all read the *same* boundary normaliser  $c_3 = 1/(8\pi)$ . This note collects the readouts. Markers: **[C]** is standard horizon physics rewritten in TFPT/seam units (no new claim) or a search target; **[E]** is an exact compiler connection (audit-level) or a reproduced numerical readout; **[O]** is a search target, not a result. Nothing here is sold as new gravitational physics; it is a *change of bookkeeping* that exposes shared structure.

**Unit conventions used throughout**

Three unit systems appear; the seam factor  $c_3 = 1/(8\pi)$  is the same number in all of them, only the carried dimensionful constants change:

| system         | convention                                                            | Hawking temperature                                                    |
|----------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|
| Planck         | $G=\hbar=c=k_B=1$                                                     | $T_H = c_3/M = 1/(8\pi M)$                                             |
| reduced Planck | $\bar{M}_{\text{Pl}} = M_{\text{Pl}}/\sqrt{8\pi},$<br>$\hbar=c=k_B=1$ | $T_H = c_3 \bar{M}_{\text{Pl}}^2/M$                                    |
| SI             | restore $G, \hbar, c, k_B$                                            | $T_H = \frac{\hbar c^3}{8\pi G k_B M} = c_3 \frac{\hbar c^3}{G k_B M}$ |

All boxed formulas below are in Planck units unless stated; the seam constant is dimensionless.

*Standard references for the formulas reframed here:* Hawking temperature and radiation (Hawking 1975); Bekenstein–Hawking entropy (Bekenstein 1973, Hawking 1975); Page time (Page 1993); fast scrambling  $t_{\text{scr}} \sim \beta \log S$  (Sekino–Susskind 2008, Maldacena–Shenker–Stanford 2016); Nariai bound (Nariai 1950); de Sitter thermodynamics (Gibbons–Hawking 1977). *Nothing below is a new derivation of these* — only a rewriting in the seam unit  $c_3$ .

## 1 The seam constant is the universal horizon temperature factor

Every Killing horizon has  $T = \hbar\kappa/(2\pi c k_B)$ . Since

$$\boxed{\frac{1}{2\pi} = 4c_3, \quad \frac{1}{8\pi} = c_3} \quad [\text{E}],$$

the universal factor *is* the seam constant:  $T_{\text{hor}} = 4c_3 \hbar\kappa/(c k_B)$ . Black holes, de Sitter and Unruh therefore share *one* thermal grammar. [C] [\[verification/v8\\_horizon.py\]](#)

## 2 Schwarzschild thermodynamics in four $c_3$ -lines

For a non-rotating, uncharged hole in Planck units ( $G=\hbar=c=k_B=1$ ):

$$\boxed{T_H = \frac{c_3}{M}, \quad S_{\text{BH}} = \frac{M^2}{2c_3}, \quad P_H = \frac{c_3}{1920 M^2}, \quad \tau_{\text{evap}} = \frac{640}{c_3} M^3} \quad [\text{C}].$$

Two of the constants are compiler numbers, not coincidences:

- the Hawking temperature factor is  $c_3 = 1/(8\pi)$  — the same P1 boundary normaliser; [E]
- the radiated-power denominator is  $1920 = |W(D_5)|$ , the Weyl-group order of the  $D_5$  carrier. [E]

*Scope of  $P_H$ :* this is the idealised massless single-channel blackbody normalisation. Greybody factors and the radiated species content shift the phenomenological evaporation coefficient; the  $c_3$ -form is the convention statement, not a general phenomenological prediction.

| object                                                  | $T_H$                           | lifetime / entropy                                                                                 |
|---------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------|
| solar-mass hole                                         | $6.17 \times 10^{-8} \text{ K}$ | $\tau_{\text{evap}} \approx 2.1 \times 10^{67} \text{ yr};$<br>$S/k_B \approx 1.05 \times 10^{77}$ |
| PBH evaporating now ( $1.7 \times 10^{11} \text{ kg}$ ) | $k_B T \approx 61 \text{ MeV}$  | relevant for $\gamma/\nu$ PBH signatures                                                           |

**The temperature is modular; the entropy carries the scalaron.** `[verification/v208_bh_thermodynamics.py]` The Hawking factor is not put in by hand: the stationary exterior modular flow  $\Delta^{it} = e^{-2\pi t K_H}$  (Bisognano–Wichmann / Tomita–Takesaki) makes the outside state KMS at inverse temperature  $2\pi$ , so  $T_H = \kappa/(2\pi)$  — and that  $2\pi$  is the seam unit  $1/(4c_3)$ , reproducing  $T_H = c_3/M$  above (the seam=horizon modular identification ties to  $[\rho, \Lambda_\Sigma] = 0$ , `[verification/v198_modular_commutator_reduction.py]`). The induced  $R + R^2$  scalaron then corrects the area law to the Wald entropy

$$S_W = \frac{f_R A}{4G} = \frac{A}{4G} \left(1 + \frac{R_h}{3M_s^2}\right), \quad M_s = c_3^{7/2} \bar{M}_{\text{Pl}},$$

an *exact* consequence of  $f(R) = R + R^2/(6M_s^2)$  (`[verification/v28_gravity_fR.py]`); the leading  $A/4G$  is the  $c_3$  area law ( $1/4 = 1/|\mu_4|$ ). **[E]** for the identities; the black-hole/modular identification is **[C]**.

### 3 Page time and scrambling

Since  $S_{BH} \propto M^2$  and  $\tau \propto M^3$ , the Page point ( $M \rightarrow M/\sqrt{2}$ ) is

$$t_{\text{Page}} = \left(1 - \frac{1}{2\sqrt{2}}\right) \tau_{\text{evap}} \approx 0.6464 \tau_{\text{evap}} \quad \text{[C]}.$$

In TFPT the Page curve is *boundary recoverability*: the physical sector is OS-reconstructed and gapped, and the recovered mutual information decays at the transfer rate,  $I(A:C|B)_n \sim \lambda_2^n = (2/3)^{6n}$ . **This is the same operator that builds the Standard Model:** the boundary transport spectrum is  $\{1, (2/3)^6, (1/3)^6\}$  and its sub-leading eigenvalue  $\lambda_2 = (2/3)^6$  is exactly the flavor-gap eigenvalue ( $\Delta_{\text{gap}} = -\log(2/3)^6 = 6 \log \frac{3}{2}$ ). One boundary transport thus governs *both* the SM flavor hierarchy and the horizon information recovery. **[E]** `[verification/v54_seam_horizon_keystones.py]`

**The recovery is a finite code** (`[verification/v221_seam_qecc.py]`). Read as a quantum channel, this is a finite recoverability (error-correction-style) code: the code space is the carrier half-spinor  $\dim S^+ = 16$ , the gapped transport  $T$  on the cusp-weight 3-space (deviation directions  $(1, -1, 0)$  and the Nariai anchor  $(1, 1, -2)$ ) is symmetric and doubly stochastic (CPTP), and on the trace-zero subspace it contracts with operator norm exactly  $(2/3)^6$ , so  $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$  — a Knill–Laflamme-type per-step recovery bound, with a free ratio breaking it. “Finite recoverability code” is **[E]**; the explicit Petz recovery map and the rank-one (one-dimensional baby-universe, cf. Engelhardt 2025) contraction limit  $\|T^n - P_\infty\| = (2/3)^{6n}$  are now realised numerically (`experiments/recovery-channel`), so the Petz identification of  $T$  with the Page recovery map is verified at the channel level **[C]**; the full holographic reading stays gated on the Seam–Horizon theorem. The scrambling prefactor carries a  $\mu_4$  signature: with  $\beta = 1/T_H = 8\pi M$ ,  $\beta/2\pi = 4M$ , so

$$\boxed{t_{\text{scr}} \sim |\mu_4| M \log S} \quad (|\mu_4| = 4) \quad \text{[E]-audit.}$$

### 4 De Sitter and the Nariai bound from the $\Lambda$ closure

With the action-grammar Hubble/ $\Lambda$  readouts ( $H_0/\bar{M}_{\text{Pl}} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$ ,  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 \sim e^{-2\alpha_\star^{-1}}$ ), the asymptotic de Sitter horizon gives

$$\boxed{\frac{T_{dS}}{\bar{M}_{\text{Pl}}} = 16c_3^2 e^{-\alpha_\star^{-1}}, \quad S_{dS} = \frac{e^{2\alpha_\star^{-1}}}{128c_3^4} = 32\pi^4 e^{2\alpha_\star^{-1}} \approx 3.32 \times 10^{122}} \quad \text{[C]}.$$

This is the area-law horizon capacity of the universe from  $(\alpha_\star^{-1}, c_3, H_\Lambda)$  alone. The maximal black hole in de Sitter (Nariai) is

$$M_N \approx 2.16 \times 10^{22} M_\odot \approx 4.3 \times 10^{52} \text{ kg}, \quad S_N = \frac{1}{3} S_{dS} \quad \text{[C]},$$

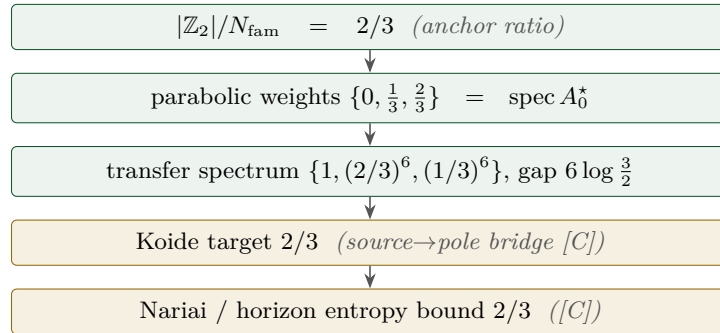
the absolute gravitational-capacity limit, not an observed structure.

## 5 The maximal black hole is the anchor: SdS in seam units

The Nariai limit is not just a capacity number — written in compiler units, the entire Schwarzschild–de Sitter (SdS) family lands on atoms that are already load-bearing elsewhere in TFPT, with no adjustable parameter on either side. [E] [verification/v101\_horizon\_anchor.py]

**The defect reading: no singularity, a seam fixed point** ([verification/v209\_bh\_regular\_core.py]).

In compiler units the interior is *not* a curvature blow-up but the seam attractor: the same gapped transport whose sub-leading eigenvalue is  $\lambda_2 = (2/3)^6$  ([verification/v56\_unique\_attractor.py]) drives  $\varphi \rightarrow \varphi_*$  ( $d\varphi/dt = 0$ ), so “ $\rho \rightarrow \infty$ ” is replaced by a fixed point; information returns through the same Page-recovery channel ( $\lambda_2$  per step, [verification/v54\_seam\_horizon\_keystones.py]); and the end state is a holographic Planck-scale floor ( $S_{BH} = A/4$ , one cell per  $|\mu_4| = 4$  Planck areas), not a point. *Honest*: a [C] structural reading — the old RN/torsion-charge metric is *not* resurrected, it is superseded by the Nariai/seam=horizon anchor here.



Green: exact motif ([E]). Gold: physical bridge ([C]) — the same ratio, two epistemic types.

**Nariai = the anchor; the Koide 2/3 is the entropy bound** [E]  
([verification/v101\_horizon\_anchor.py])

With  $f(r) = 1 - 2M/r - \Lambda r^2/3$ , the horizon cubic at the maximal mass  $M_N = 1/(3\sqrt{\Lambda}) = 1/(N_{\text{fam}}\sqrt{\Lambda})$  is, in units  $1/\sqrt{\Lambda}$ ,

$$t^3 - 3t + 2 = (t-1)^2(t+2) \quad \text{roots } (1, 1, -2),$$

with coefficients  $(-3, 2) = (-N_{\text{fam}}, |\mathbb{Z}_2|)$  — and the root pattern is exactly the **traceless projection of the anchor**:  $a - \frac{p_1(a)}{3}\mathbf{1} = -\frac{1}{3}(1, 1, -2)$  for  $a = (1, 1, 2)$  (compare  $\chi_a = (t-1)^2(t-2)$ , the anchor characteristic polynomial). Five exact companions:

- **Entropy bound = the Koide branch value.** Each Nariai horizon carries  $S_{dS}/3 = S_{dS}/N_{\text{fam}}$  and the total is  $S_{\text{Nariai}}/S_{dS} = 2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$  (deficit exactly  $S_{dS}/3$ ). The interpolation in  $x = r_b/r_c$  is the closed form  $S_{\text{tot}}/S_{dS} = (x^2+1)/(x^2+x+1)$  — denominator =  $\Phi_3(x)$ , the  $N_{\text{fam}}$  cyclotomic — monotone from 1 to  $2/3$  with its minimum exactly at the merge (Fig. 2a). The floor admits a *variation-free* one-line proof:  $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ , so  $S_{\text{tot}} \geq \frac{2}{3}S_{dS}$  with equality *iff*  $x = 1$  (the Nariai merge) [E] ([verification/v190\_nariai\_entropy\_bound.py]).
- **Three-sheet conservation.** The cubic is traceless with fixed  $e_2$ , so  $\pi \sum_i r_i^2 = 6\pi/\Lambda = |\mathbb{Z}_2| S_{dS}$  for *every*  $M$ : the black hole redistributes entropy between the two physical sheets and the virtual third root; the three-sheet total never moves.
- **Koide-form quotient of the roots.**  $Q_{\text{geom}} = \sum r_i^2 / (\sum |r_i|)^2$  runs monotonically over  $[\frac{3}{8}, \frac{1}{2}]$ : pure dS gives  $\frac{1}{2} = |\mathbb{Z}_2|/|\mu_4| = \delta$  (the half-step), Nariai gives  $\frac{3}{8} = p_2(a)/e_1(a)^2$  — exactly the two nonzero  $SU(4)_1$  conformal weights (Fig. 2b).
- **The mass line is a double cover.**  $\text{disc}_r \propto (1-3m)(1+3m)$  in  $m = M\sqrt{\Lambda}$ : branch points  $m = \pm 1/N_{\text{fam}}$ , separation  $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$ , deck involution = the black-hole  $\leftrightarrow$  cosmological horizon swap — the gravitational twin of the flavor sheet  $\mathbb{Z}_2$  and of the split  $N_{\text{fam}}$ -linear cover  $(3x+2)(3x+5)$  of the flavor sector (Fig. 1). The two branch structures share one  $N_{\text{fam}}$ -linear ramification: the affine map  $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$  carries the SdS branch points  $m = \pm 1/N_{\text{fam}}$  onto the flavor branch labels  $\{2, 5\} = \{|\mathbb{Z}_2|, g_{\text{car}}\}$  (midpoint  $7/2$ ). This is an exact *relabeling* — the shared invariant is the  $2/3$  ramification, not the map, which exists for any two pairs of points — so it is recorded as

a [C] structural alignment, not a theorem ([verification/v191\_universal\_branch\_line.py]).

- **Temperature lemma / orientation.**  $|\kappa_b/\kappa_c| = (2x+1)/(x(x+2)) > 1$  for  $x < 1$  (equality iff  $x = 1$ ): the black-hole sheet is *always* hotter, so evaporation flows *away* from the merge — the anchor configuration is the **repeller** and the democratic endpoint the attractor, the same orientation as the flavor flow (Koide attractor / carrier repeller, the frontier note). *Audit reading.*

Completing the seam-unit table:  $dM = c_3 \kappa dA$  (first law),  $M = 2c_3 \kappa A$  (Smarr),  $S \leq ER/(4c_3)$  (Bekenstein),  $P_H = c_3/(1920 M^2)$ , lifetime  $\tau = 5120\pi M^3 = 128 g_{\text{car}} M^3/c_3$  ( $128 = 2^7$  the dS prefactor, 7 the scalaron exponent; photon/geometric-optics conventions, typed external), Kerr  $A_{\text{ext}} = M^2/c_3$  with  $A_{\text{ext}}/A_{\text{Schw}} = 1/2 = 1/|\mathbb{Z}_2|$  and  $S_{\text{ext}} = M^2/(4c_3)$ , and the Bekenstein–Mukhanov–Hod area quantum  $4 \ln 3 = \ln 81 = \ln(N_{\text{fam}}^4) = \ln(\text{disc of the flavor cover})$  ([C] as the QNM reading). *Honest scope:* the GR identities are textbook facts; the [E] content is six independent landings on already-load-bearing atoms with zero free parameters. The “carrier in the bulk” reading (two glue chiralities = two horizon sheets, ramification = Narai) stays [C]. No new data test ( $M_N$  today  $\sim 10^{22} M_\odot$ ): a structure surface, not a measurement surface.

**The dual anchor: the inverse flavor response is the Narai root [E]**  
([verification/v134\_dual\_anchor.py])

The Narai pattern is not only the traceless anchor — it is stored *inside the flavor compiler* as a dual invariant. For the residue/winding operators  $R, L$  of the flavor sector and the anchor  $a = (1, 1, 2)$ ,

$$d := a^\top R^{-1} = a^\top L^{-1} = \left(-\frac{1}{2}, -\frac{1}{2}, 1\right), \quad d \cdot \mathbf{1} = 0, \quad d \cdot a = 1, \quad (1, 1, -2) = -2d,$$

i.e.  $d = \frac{3}{2}(a - \frac{4}{3}\mathbf{1})$  is the *normalised traceless anchor*, proportional to the Narai double-root pattern above. The invariance is exact and structural (Sherman–Morrison): with  $L = R + 6\mathbf{1}e_1^\top$ , a covector is winding-invariant iff it annihilates  $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$  — the winding deformation lives in the democratic direction  $\mathbf{1}$ , which  $d$  kills. Controls:  $\mathbf{1}, e_1$  and the torsion normal  $n = (5, -9, 6)$  do *not* lie on the invariance plane  $(\frac{1}{4}, \frac{1}{4}, -\frac{5}{2})$ , so the anchor’s membership is special. Together  $(d, n)$  form the dual normal pair of the flavor boundary:  $d$  reads the traceless horizon structure,  $n$  reads first-generation torsion ( $n^\top R = (8, 0, 0)$ ,  $n^\top L = (20, 0, 0)$ ). *Typing:* all identities [E]; the reading — a third, purely *algebraic* leg of the flavor  $\leftrightarrow$  horizon bridge, beside the shared clock spectrum ([verification/v126\_clock\_wall\_bridge.py]) and the entropy power law ([verification/v129\_entropy\_power\_law.py]) — stays [C].

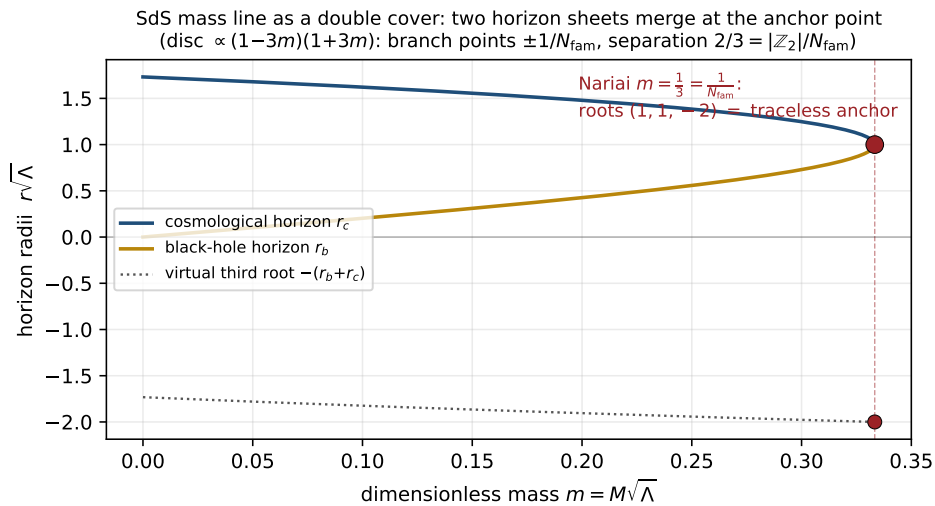


Figure 1: The SdS mass line as a double cover: black-hole and cosmological horizon sheets merge at the Narai point  $m = 1/N_{\text{fam}}$ , where the three roots form the traceless anchor pattern  $(1, 1, -2)$ . [verification/v101\_horizon\_anchor.py]

**One orientation: the anchor is the *stationary repeller* in both sectors ([E]/[C]).** The evaporation direction of the temperature lemma upgrades to exact variational structure on both





Figure 2: (a) Total SdS entropy interpolates from  $S_{dS}$  to the Nariai bound  $\frac{2}{3}S_{dS}$  via  $(x^2+1)/\Phi_3(x)$ . (b) The Koide-form quotient of the three roots runs exactly over the nonzero  $SU(4)_1$  weights  $[\frac{3}{8}, \frac{1}{2}]$ . [\[verification/v101\\_horizon\\_anchor.py\]](#)

Twin double covers: the same split  $N_{\text{fam}}$ -linear form in flavor and in classical gravity

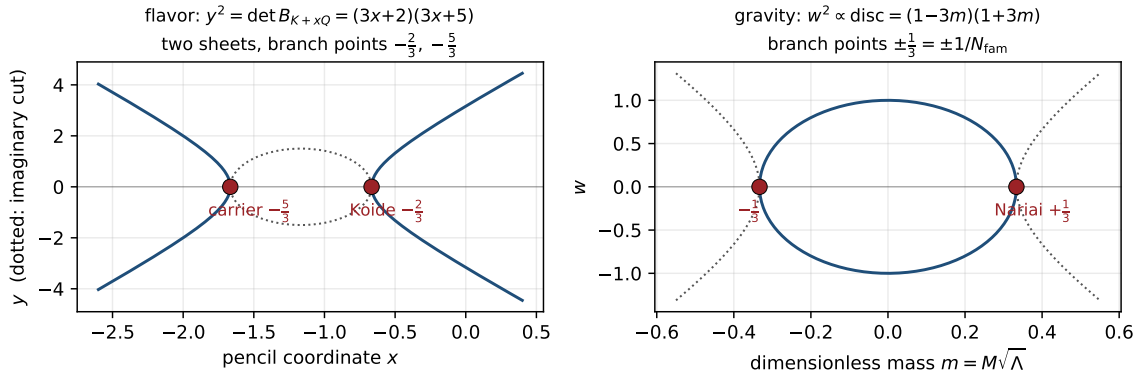


Figure 3: Twin double covers: the flavor anchor-block cover  $y^2 = (3x+2)(3x+5)$  and the SdS mass-line cover  $w^2 \propto (1-3m)(1+3m)$  share the split  $N_{\text{fam}}$ -linear form with rational branch points at spine-over-family values. [\[verification/v101\\_horizon\\_anchor.py\]](#)

one-dimensional moduli lines (Fig. 4). **Flavor:** the canonical relaxation  $\dot{q} = (\Delta/N_{\text{fam}})(q-2)(q-5)$  is the *gradient flow* of the cubic potential

$$V(q) = -\frac{\Delta}{N_{\text{fam}}} \left( \frac{q^3}{3} - \frac{7}{2}q^2 + 10q \right), \quad \boxed{V''(2) = +\Delta, \quad V''(5) = -\Delta}$$

— the critical points are exactly the two branch points, the stationary curvatures are  $\pm$ the transfer gap, the inflection sits at  $q = \frac{7}{2} = \text{scalaron}/2$ , and the Lyapunov rate is constant:  $d(-\ln \rho)/dt = \Delta$  exactly. **Gravity:** the SdS entropy functional has

$$\frac{d}{dx} \frac{S_{\text{tot}}}{S_{dS}} = \frac{(x-1)(x+1)}{\Phi_3(x)^2}, \quad \boxed{\left( \frac{S_{\text{tot}}}{S_{dS}} \right)''(1) = \frac{2}{9} = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}^2} = \frac{2}{3} \cdot \frac{1}{3}}$$

— the Nariai/anchor point is the *unique* physical stationary point, its curvature is a grammar constant (= the product of the two Nariai entropy fractions), and the temperature lemma makes the evaporation an entropy *ascent* away from it. **Orientation theorem [E]:** in both sectors the anchor-type configuration is a stationary point of the natural functional and the dynamics

flows away from it, with grammar-constant stationary curvatures. *Honest disanalogies (recorded)*: the flavor attractor is itself a branch point while the gravity attractor is the smooth endpoint; the flavor functional is the potential of a [C]-typed relaxation while the gravity functional is the Bekenstein–Hawking entropy — “one variational principle of the seam” stays a *reading* [C], and no open gate is touched. [verification/v102\_seam\_orientation.py]

One orientation: the anchor is the stationary repeller in both sectors

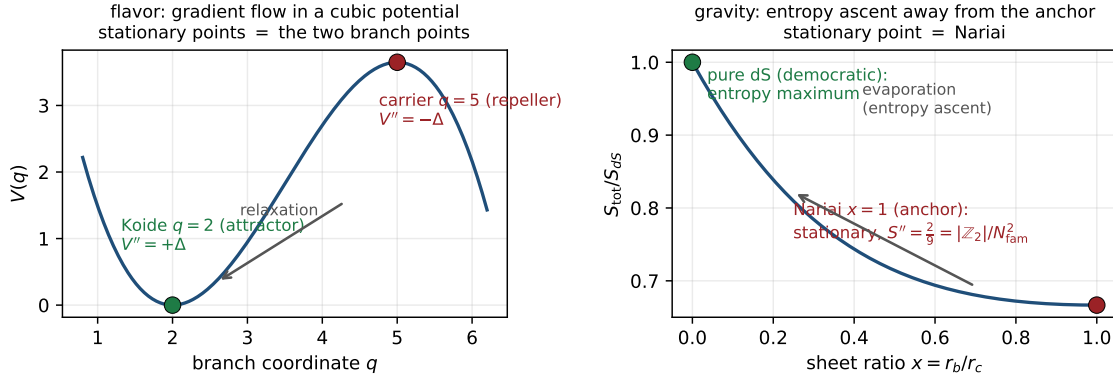


Figure 4: The orientation theorem: the anchor configuration is the stationary repeller of the natural functional in both sectors — flavor (cubic potential, curvatures  $\pm\Delta$ ) and gravity (SdS entropy, curvature  $2/9 = |\mathbb{Z}_2|/N_{\text{fam}}^2$ ). [verification/v102\_seam\_orientation.py]

**The trisection normal form: the canonical coordinate exists [E]**  
([verification/v103\_trisection\_normal\_form.py])

The curvature-matching question of the previous box has a closed answer. The SdS horizon cubic is uniformized by **angle trisection**: with  $r = 2 \cos \theta$  (units  $1/\sqrt{\Lambda}$ ),  $r^3 - 3r = 2 \cos 3\theta$  *exactly*, so the cubic is  $\cos 3\theta = -3m$  — the three horizon roots are the three trisection branches (the  $\mathbb{Z}_3$  rotation = the triality deck, cf. coker  $Q = \mathbb{Z}/N_{\text{fam}}$ ). In the centered angle  $\psi$  ( $\cos \varphi = -3m$ ,  $\psi = \varphi - \pi$ ; Nariai  $\psi = 0$ , pure dS  $\psi = \pi/2$ , deck  $\psi \rightarrow -\psi$ ):

$$m = \frac{\cos \psi}{N_{\text{fam}}}, \quad \frac{S_{\text{tot}}}{S_{\text{dS}}} = \frac{4}{3} - \frac{2}{3} \cos \frac{2\psi}{3}$$

— *one* cosine whose mean ( $|\mu_4|/N_{\text{fam}}$ ), amplitude and frequency (both  $|\mathbb{Z}_2|/N_{\text{fam}}$ ) are glue atoms over the family count (Fig. 5). Three exact consequences:

- **Canonical curvature = the Koide constant to the family power:**  $\sigma''(\psi=0) = \frac{8}{27} = (\frac{2}{3})^3 = (|\mathbb{Z}_2|/N_{\text{fam}})^{N_{\text{fam}}}$ .
- **Invariant base slope:**  $d\sigma/dm|_{\text{Nariai}} = -\frac{8}{9} = -\text{rank } E_8/N_{\text{fam}}^2$  (coordinate-invariant, cross-checked in both coordinates; generally  $-\frac{4}{3} \sin(2\psi/3)/\sin \psi$ ); dimensionful  $dS_{\text{tot}}/dM|_N = -r_N/(N_{\text{fam}} c_3)$ .
- **The bridge:** the flavor invariant is a *rate*, multiplier  $(2/3)^6 = (2/3)^{2N_{\text{fam}}}$  per transport step; the gravity invariant is a *curvature*,  $(2/3)^{N_{\text{fam}}}$  in the canonical angle — same base  $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$ , exponent ratio  $2 = |\mathbb{Z}_2|$ . What is still missing for a full identification is the gravity-side *clock* (the near-Nariai evaporation generator that converts curvature into a rate) — the same class of missing object as the Koide transfer generator (frontier note): **one clock, two known geometries**. That identification stays [C].

**The classical clock speaks anchor — and the honest  $(2/3)$ -test ([E]/[C]).** The clock question of the previous box has a *classical* half, and it is pure GR (Ginsparg–Perry 1983; Bousso–Hawking class): linearize around the Nariai geometry  $dS_2 \times S^2$ . **Static pin [E]:** on the  $dS_2$  static patch the function  $\phi(\rho) = \rho$  satisfies  $\frac{d}{d\rho}[(1 - \Lambda\rho^2)\phi'] = -2\Lambda\phi$  *exactly* — and this static mode *is* the exact SdS



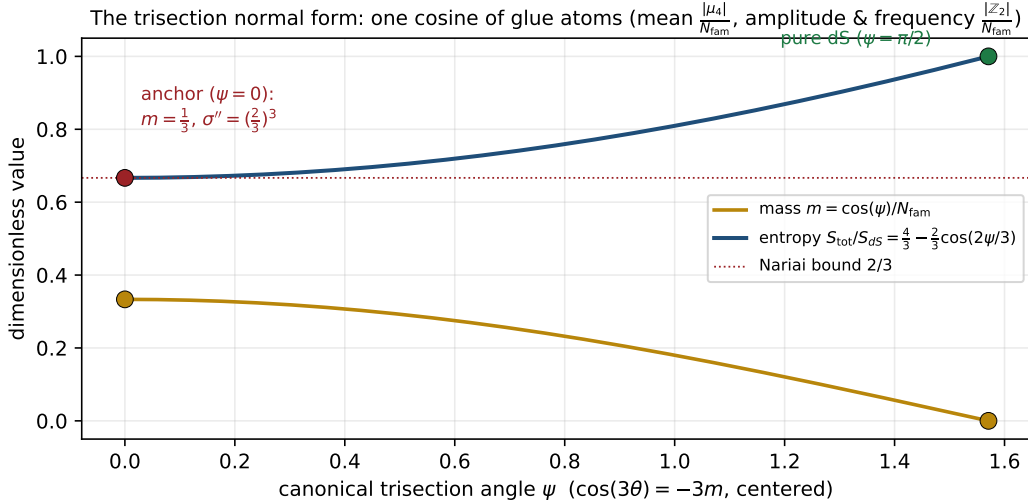


Figure 5: The trisection normal form: in the canonical angle  $\psi$  the mass and the total entropy are pure (co)sines built from glue atoms; the anchor sits at  $\psi = 0$  with canonical curvature  $(2/3)^3$ .  
[\[verification/v103\\_trisection\\_normal\\_form.py\]](#)

two-horizon deformation, so the  $S^2$ -radius modulus mass is pinned internally:  $m^2 = -2\Lambda = -|\mathbb{Z}_2|\Lambda$  (the Laplace-type form of the linearization is the one external GR import, typed honestly). The Euclidean spectrum is the Ginsparg–Perry tower  $(l(l+1) - 2)\Lambda$ : exactly *one* negative mode  $-|\mathbb{Z}_2|\Lambda$ . **The clock [E]:** in Hubble units the homogeneous modes  $\phi \sim e^{\lambda H t}$  obey

$$\chi_{\text{clock}}(\lambda) = \lambda^2 + \lambda - 2 = (\lambda - 1)(\lambda + 2)$$

— **the anchor quadratic:** eigenvalues  $\{1, -2\}$  = the distinct anchor roots, and the Nariai cubic factors as  $(t - 1)^2(t + 2) = (t - 1) \cdot \chi_{\text{clock}}(t)$ . The anchor thus appears a *third* time: as configuration roots, as the curvature base, and as the spectrum of the linearized clock. The horizon split is linear in the canonical angle with coefficient  $1/\sqrt{N_{\text{fam}}}$ , and the entropy deviation grows at exactly  $2H = |\mathbb{Z}_2| \times \text{Hubble}$ . **Honest (2/3)-test [C]:** the classical clock eigenvalues are *integers*, not (2/3)-powers — the (2/3)-powers stay in the functional (curvature  $(2/3)^{N_{\text{fam}}}$ ). The conversion curvature  $\rightarrow$  rate at one loop (quantum (anti-)evaporation) remains external QFT input: the bridge to the flavor rate  $(2/3)^{2N_{\text{fam}}}$  is *not* closed here. The seam variational principle now has exactly *one* missing object: the quantum clock. [\[verification/v104\\_nariai\\_clock.py\]](#)

**The quantum-clock target, made quantitative ([E]/[C]).** Instead of guessing the quantum clock, its job description can be pinned exactly. **(1) The classical clock already lives on the cusp ladder [E]:** across the physical Ginsparg–Perry modes ( $l = 0$ :  $(\lambda - 1)(\lambda + 2)$ ;  $l = 1$ :  $\lambda(\lambda + 1)$ ) the decay exponents are

$$\{0, -1, -2\} = -\text{Spec}(V) = -N_{\text{fam}} \times \text{cusp weights}$$

— exactly the grading that carries the transfer operator’s eigenspaces (Papers 2/6); the  $l \geq 2$  modes are damped oscillations with  $\text{Re } \lambda = -\frac{1}{2} = -\delta$  (audit). **(2) The exact target [E]:** the transfer rates  $\{0, \Delta, 6 \ln 3\}$  are *not* weight-linear —  $\text{rate}(2)/\text{rate}(1) = \log_{3/2} 3 = 1 + \log_{3/2} 2 = 2.7095$ , with the exact identity  $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$ : *the quantum clock must bend the weight-linear spectrum by  $\log_{9/4} 3 = 1.35476$  per weight step.* **(3) Coupling landscape [E]:** the dimensionless clock coupling  $\kappa = (c/24\pi)(\Lambda/\bar{M}_{\text{Pl}}^2)$  with  $c = 8$  is  $7.6 \times 10^{-122}$  at the cosmological Nariai (hopeless) and  $\kappa = 1/(3\pi) = 0.106$  at seam curvature — the only regime where an  $O(0.35)$  bend is reachable, and it is borderline non-perturbative: one-loop linear response cannot fix the bend; an  $O(1)$  resummation or an exact seam construction is required. *Status [C]:* R1 is sharpened from “find a

clock” to “produce the bend  $\log_{3/2} 3$  on the cusp ladder at coupling  $O(1/3\pi)$ ”. Nothing is fitted; the identification stays open. [\[verification/v107\\_quantum\\_clock\\_target.py\]](#)

**The resummed clock: the bend has a closed form ([E]/[C]).** The “bend” is not an exotic target — the frozen transfer spectrum *is* a resummed logarithm. **(1) Spectrum reading [E]:** the eigenvalues  $\{1, (\frac{2}{3})^6, (\frac{1}{3})^6\}$  are exactly

$$\lambda_n = \left(1 - \frac{n}{N_{\text{fam}}}\right)^{p_2}, \quad n = 0, 1, 2 \quad (p_2 = 6 = |R^+(A_3)|),$$

so the clock has the closed form  $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$  — it reproduces  $\{0, \Delta, 6 \ln 3\}$ , and the [v107](#) bend  $\log_{3/2} 3$  becomes a one-line consequence. **(2) The linearisation carries the sheet [E]:**  $\text{rate}(n) = |\mathbb{Z}_2| n + \frac{n^2}{3} + \frac{2n^3}{27} + \dots$  — the one-loop term has slope exactly  $|\mathbb{Z}_2| = 2$  relative to the *integer* classical Ginsparg–Perry clock  $\{0, -1, -2\}$ : the quantum clock is the sheet-doubled classical clock plus the geometric tail  $\sum_k (n/N)^k/k$ . **(3) The wall at  $N_{\text{fam}}$  [E]:**  $\text{rate} \rightarrow \infty$  as  $n \rightarrow N_{\text{fam}}$  — the resummed logarithm has a pole at the family count, so the cusp ladder *must* end at  $n = N_{\text{fam}} - 1 = 2$ : the three-level transfer spectrum is forced by the clock itself. *R1 re-sharpened [C]:* the semiclassical job is now “derive  $-p_2 \ln(1 - n/N_{\text{fam}})$  from the near-Nariai one-loop at  $\kappa = 1/(3\pi)$ ” — a closed-form target, with the linear-response slope  $|\mathbb{Z}_2|$  as its first checkpoint. All spectra frozen ([v54/v69/v95/v104/v107](#)); nothing fitted. [\[verification/v124\\_resummed\\_clock.py\]](#)

**The clock and the wall share a spectrum ([E]/[C]).** Both ends of the resummed clock connect to established exact objects. **(1) The weights are the parabolic weights [E]:** the arguments  $n/N_{\text{fam}} = \{0, \frac{1}{3}, \frac{2}{3}\}$  of the resummed logarithm are *exactly*  $\text{spec } A_0^*$  — the spectrum of the exact ( $U_{\text{wall}}$ ) residue (research contracts):

$$\lambda_n = (1 - \alpha_n)^{p_2}, \quad \alpha_n \in \text{spec } A_0^* = \{0, \frac{1}{3}, \frac{2}{3}\}$$

— the horizon clock and the flavor wall share their spectrum through the complement map  $\alpha \mapsto 1 - \alpha$ . **(2) Checkpoint 1 passes classically [E]:** the clock’s linear coefficient  $p_2/N_{\text{fam}} = 2 = |\mathbb{Z}_2|$  *equals* the established classical entropy-deviation rate  $2H = |\mathbb{Z}_2| \times \text{Hubble}$  ([v104](#)) — the slope-2 linear response is already the classical entropy rate; the genuinely *quantum* content of R1 isolates to the geometric tail  $\sum_{k \geq 2} \alpha^k/k$ . **(3) The determinant is the entropy curvature [E]:**  $\det(\mathbf{1} - A_0^*) = \frac{2}{9} = |\mathbb{Z}_2|/N_{\text{fam}}^2$  — exactly the gravity entropy curvature at the anchor ([v102](#)), a fourth independent appearance;  $\text{tr}(\mathbf{1} - A_0^*) = 2 = |\mathbb{Z}_2|$ ,  $\det A_0^* = 0$ . *R1 re-targeted [C]:* derive “rate  $= -p_2 \ln(1 - \alpha)$ ” with  $\alpha$  the Mehta–Seshadri parabolic weight — first order classical, tail quantum: **one spectrum, two gates, one object  $A_0^*$** . [\[verification/v126\\_clock\\_wall\\_bridge.py\]](#)

**The clock is a log-determinant: the tail is the ring (RPA) resummation ([E]/[C]).** The “ $O(1)$  resummation” demanded by [v107](#) is now *typed*. For a rank-one insertion  $P$  ( $\text{tr } P^k = 1$ ),  $-\ln \det(\mathbf{1} - \alpha P) = -\ln(1 - \alpha) = \sum_k \alpha^k/k$ , so the resummed clock is exactly

$$\text{rate} = -p_2 \text{Tr} \ln(\mathbf{1} - \alpha P)$$

—  $p_2 = 6$  *identical* one-loop ring (RPA) towers, **one per hexagon site** (= positive root of  $A_3$ ), with coupling  $\alpha$  = the parabolic weight. *Ring ladder [E]:* the  $k=1$  ring is the *classical* term ( $6\alpha = |\mathbb{Z}_2|n$ , the [v104/v126](#) entropy rate); the  $k=2$  ring is the first quantum correction ( $\frac{1}{3}$  at weight 1); the partial sums converge to  $\{\Delta, 6 \ln 3\}$  monotonically from below. *Coupling cross-link [E] (audit):* the established seam coupling is  $\kappa = \frac{c}{24\pi} = \frac{8}{24\pi} = \frac{1}{3\pi} = \alpha_1/\pi$  — the *first parabolic weight over  $\pi$* . *The wall [E]:* the ring series diverges at  $\alpha \rightarrow 1$  ( $n \rightarrow N_{\text{fam}}$ ) — the [v124](#) wall is the standard RPA instability. *R1 final form [C]:* show that the near-Nariai one-loop effective action of the seam mode, with insertion = parabolic weight and multiplicity  $p_2$ , is this ring sum — “**the seam clock is the RPA determinant of one mode on six sites**”. The resummation type is identified; the derivation itself remains open.

Where the six sites live ([\[verification/v128\\_graded\\_hull.py\]](#)): the Ginsparg–Perry  $l = 1$  sector — the

zero modes  $\lambda \in \{0, -1\}$  of the classical clock (v104/v107) — has  $2l+1 = 3$  modes, and sheet doubling gives  $2 \times 3 = 6 = p_2$  [E]: the ring multiplicity equals the sheet-doubled zero-mode count. [C] reading (recorded): zero modes are exactly the modes that force nonperturbative resummation in de Sitter one-loop physics — the natural home of the six RPA towers. [verification/v127\_ring\_resummation.py]

**The clock is an entropy power law** ([E]/[C]). What the resummation argument  $1 - n/N_{\text{fam}}$  is physically: the three established Nariai entropy levels (v101) are  $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$  (pure dS; two-horizon total; single horizon — deficit  $S_{dS}/3$  per step) — exactly the complementary cusp weights. Hence [E]:

$$\lambda_n = \left( \frac{S_n}{S_{dS}} \right)^{p_2}, \quad \text{rate}(n) = p_2 \ln \frac{S_{dS}}{S_n}$$

— the transfer eigenvalues are the *entropy fractions to the hexagon power*; the rates are hexagon-weighted log-entropy deficits. *The coupling is the entropy deficit* [E] (*triple identity*):  $\alpha = n/N_{\text{fam}} =$  the Mehta–Seshadri parabolic weight (v115/v126)  $= 1 - S_n/S_{dS} = \Delta S_n/S_{dS}$  — the RPA insertion strength is the fractional entropy deficit; the coupling is not a free parameter in *any* of its three readings. *Orientation* [E]: weight  $0 \leftrightarrow$  pure dS (maximal entropy, stationary attractor), weight  $2 \leftrightarrow$  single horizon (fastest decay) — the established repeller  $\rightarrow$  attractor flow. *R1, thermodynamic form* [C]: derive  $\Gamma_n \propto (S_n/S_{dS})^{p_2}$  from the near-Nariai one-loop — a Gibbons–Hawking-type entropy power law with exponent = the hexagon size (audit:  $p_2 = 6 = 2h$  reads as the two-point form of a weight  $h = 3 = N_{\text{fam}}$  operator; recorded, not claimed).

*The Born square* ([verification/v130\_born\_square.py]): the weight audit is now *derived* from mode counting [E]. In the standard collective-coordinate treatment each zero mode contributes a moduli factor  $\propto S^{1/2}$ , so with  $n_{\text{zero}} = 6$  the transition *amplitude* scales as  $A \sim (S/S_{dS})^3$  and the *probability* is its Born square  $\Gamma = |A|^2 = (S/S_{dS})^6$ :

$$h = \frac{n_{\text{zero}}}{2} = 3 = N_{\text{fam}}, \quad p_2 = 2h \text{ (the Born rule)}$$

— and the 3 is the dimension of the  $l=1$  eigenspace = the number of *proper conformal Killing vectors* of  $S^2$  (the classic near-Nariai zero modes, Ginsparg–Perry/Bousso–Hawking class), while the  $\times 2$  is the *horizon pair* = the SdS mass-line deck (v101). The same 3 counts the generations and the sphere’s conformal directions. *Residue* [C]: justify the per-zero-mode  $S^{1/2}$  normalisation on the Nariai instanton for exactly these six modes — a textbook-class computation; no exotic object remains in R1.

*The measure is the area* ([verification/v131\_measure\_is\_area.py]): the geometric core of that normalisation is exact [E]: on  $S^2(r)$  the  $L^2$  norm of *each* of the three normalised  $l=1$  harmonics is

$$\|Y_{1m}\|^2 = r^2 = \frac{A}{4\pi}$$

(all three integrated exactly) — the zero-mode norm *is* the horizon area, hence the entropy. The collective-coordinate Jacobian per mode is the mode norm  $\sim S^{1/2}$  (standard change of variables), so: norm = area  $\rightarrow$  Jacobian  $= S^{1/2} \rightarrow$  six modes  $\rightarrow$  amplitude  $S^3 \rightarrow$  Born  $\rightarrow S^6$  — every exponent factor carries a derivation-level origin. The  $4\pi$  is the same Gauss–Bonnet primitive that builds  $c_3 = \frac{1}{2}(4\pi)^{-1}$ , and it *cancels* in all ratios — exactly why the clock sees only entropy fractions. *The last step* [C]: show  $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$  (non-zero-mode determinant cancellation; same local near-Nariai geometry). [verification/v129\_entropy\_power\_law.py]

*The cancellation, exact within the family* ([verification/v144\_detratio\_family.py]): that step is now *derived* in the within-family sense [E]. The traceless SdS cubic forces

$$r_b r_c = 1 - \frac{\Delta^2}{3} \quad (\Delta = r_b - r_c \text{ the horizon split; exact, no higher corrections})$$

( $e_2$ -rigidity), and with the v132 anomaly  $\zeta(0)|_{\det'} = -\frac{2}{3}$  per sphere the two-sphere non-zero-mode determinant ratio against Nariai is the closed form  $\det'_n / \det'_N = (1 - \Delta^2/3)^{4/3}$ : **no first-order term in the split** — the hoped-for cancellation is a consequence of  $e_2$ -rigidity plus

the established anomaly, not an assumption. In deficit form,  $\varepsilon = 1 - c = \frac{3}{8}\Delta^2 + O(\Delta^4)$  gives  $\det'_n / \det'_N = 1 - \frac{32}{27}\varepsilon + O(\varepsilon^2)$  with  $\frac{32}{27} = |\mu_4|(\frac{|\mathbb{Z}_2|}{N_{\text{fam}}})^3$  (the **v103** canonical curvature times the glue order; audit-typed). *Honest scope [C]*: the finite-weight comparison ( $S_n/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ ) still runs through the **v132/v133** budget — a naive global-rescaling reading would shift the Born exponent  $6 \rightarrow \frac{14}{3}$  (not an atom), so the absorption must proceed via the Hubble-unit conversion; the frozen transfer spectrum is untouched, and the remaining [C] now has its obstruction stated sharply.

*The clock is a Gaussian zero-mode integral ([verification/v147\_clock\_gaussian.py])*: the **v127** typing job is discharged at the model level, and the bend is  $\det'$ -clean [E]. For  $p_2$  Gaussian zero modes whose per-mode variance scales by  $(1 - \alpha)$  — forced by norm = area (**v131**) — the Born-squared partition ratio is exactly  $(1 - \alpha)^{p_2}$ , i.e. rate =  $-p_2 \ln(1 - \alpha)$ , and the  $\ln$  series is the ring sum  $p_2 \sum_k \alpha^k/k$  term by term. Sharper still: the two quantum weights  $n = 1, 2$  are bookkeeping levels of one Euclidean geometry (Nariai  $S^2 \times S^2$ , **v101**), so  $\det'_1 = \det'_2$  identically — **the bend**  $\log_{3/2} 3$  (**the v107 target**) **carries no determinant correction at all**. The single across-topology step is the  $\alpha = 0$  reference (round dS vs Nariai), exactly where the **v132/v133** budgets and the **v144** obstruction live; its classical side is already matched (**v126**: slope  $2 = |\mathbb{Z}_2|$ ). *What stays [C]*: the measure identification (“the gravitational near-Nariai zero-mode measure is this Gaussian model”) and the reference normalisation — structure, spectrum, bend and ring typing are exact.

**The det-ratio anomaly is the Koide constant ([E]/[C])**. The det-ratio question acquires its first *exact* invariant. The Ginsparg–Perry sphere operator on the unit  $S^2$  is  $L = -\Delta - 2$  (the  $-2 = -|\mathbb{Z}_2|$  is the **v104** modulus mass), with eigenvalues  $(l+2)(l-1)$  and kernel the three  $l=1$  zero modes. Its heat coefficient is made of atoms [E]:  $a_1 = \frac{R}{6} - E = \frac{1}{3} + 2 = \frac{1}{N_{\text{fam}}} + |\mathbb{Z}_2| = \frac{7}{3}$ , and the scaling anomaly of the non-zero-mode determinant is

$$\zeta(0)|_{\det'} = a_1 - \dim \ker = \frac{7}{3} - 3 = -\frac{2}{3} = -\frac{|\mathbb{Z}_2|}{N_{\text{fam}}}$$

— confirmed independently by numerical continuation of the heat trace ( $\sim 10^{-7}$ ): **the det-ratio anomaly is the Koide constant**, the eighth appearance of  $2/3$  and the first as a *spectral* anomaly. *Consequence [E]*:  $\det'(cL) = c^{\zeta(0)} \det'(L)$  — the determinant ratio between configurations is *not* automatically 1; the  $S^2$ -sector share of the scaling budget is exactly  $-\frac{2}{3}$  per Hubble-unit conversion, while the dimensionless tower itself is configuration-independent (pure-number GP spectrum). *The budget statement [C] (the remaining step, made precise)*: show the *full*  $\zeta(0)$  budget of the 4d Nariai fluctuation problem ( $dS_2$  + gauge/ghost + this  $S^2$  sector) vanishes or is absorbed into the established power law — the  $S^2$  share is pinned exactly; nothing is claimed about the total. [verification/v132\_detratio\_anomaly.py]

**The zeta budget: the reduced route carries the anchor ratios ([E]/[C])**. Euclidean Nariai is  $S^2 \times S^2$  — the temporal sector continues to the *same* unit sphere, so both candidate determinant routes are exactly computable. (i) *The reduced route [E]*: each 2d sector carries the **v132** operator with  $\zeta(0)|_{\det'} = -\frac{2}{3}$ ; the two sectors give

$$\text{budget}_{\text{red}} = -\frac{2}{3} - \frac{2}{3} = -\frac{4}{3} = -\frac{e_1}{p_0} \text{ (minus the seed gain)}$$

— the per-sector share and the total are *two of the three v119 anchor-ratio triad values*; the six zero modes split  $3 + 3$  across the two spheres: the sheet doubling is the two-sphere structure of Euclidean Nariai (= the horizon pair, **v101**). (ii) *The naive 4d route [E]*: with the standard  $S^2$  heat expansion  $1/t + \frac{1}{3} + t/15$  and the symmetric split,  $a_2(4d) = (\frac{4}{3})^2 + 2 \cdot \frac{9}{10} = \frac{161}{45}$ , six zero modes,  $\zeta(0)|_{\det', 4d} = \frac{161}{45} - 6 = -\frac{109}{45}$  — *no* anchor-atom reading. (iii) *Route discrimination [C]*: the budget computation structurally favours the **reduced seam reading** (pure anchor-ratio anomalies), consistent with the theory’s 2d-seam architecture throughout — route-selection evidence, not proof. *Remaining [C]*: the graviton/ghost shares on  $S^2 \times S^2$  (Volkov–Wipf class, external standard) — a finite list of known heat coefficients. [verification/v133\_zeta\_budget.py]

**The Volkov–Wipf firewall: the external edge is two copies of the reduced budget** ([E]/[C]). The “finite list” above is now *filled in externally*. For the full one-loop graviton path integral on  $S^2 \times S^2$  the log-coefficient is (arXiv:2506.02142, Table 4.1, agreeing with Volkov–Wipf, Nucl. Phys. B582 (2000) 313):

$$\alpha_{\text{bulk}} = \frac{22}{45}, \quad \alpha_{\text{edge}} = -\frac{8}{3}, \quad \alpha_{\text{PI}} = -\frac{98}{45}$$

(no anchor atom; the difference to the reduced budget is  $-\frac{38}{45}$  with  $38 = 2 \cdot 19$ ). The structural finding: the external *edge* factor consists of **two identical copies** (one per horizon), and

$$\alpha_{\text{edge}} = -\frac{8}{3} = 2 \times \left(-\frac{4}{3}\right)$$

— the per-copy magnitude  $\frac{4}{3} = e_1/p_0$  is *exactly* the reduced seam budget above (the seed gain;  $-\frac{2}{3}$  per sector). The TFPT reduced route lands on the **edge sector** of the external bulk/edge split; the bulk ( $\frac{22}{45}$ ) is the ideal graviton gas, which the seam clock does not contain; two copies = the horizon pair = the sheet doubling (v101; the 3+3 zero-mode split). *Typing decision* [C] (recorded): R1 = the reduced zero-mode/RPA seam clock = an edge-sector object; the full 4d Volkov–Wipf determinant is the *gravity firewall* (its  $-\frac{98}{45}$  is absorbed in the  $\mu_0/(\Lambda G)^3$  normalisation), not a TFPT atom. Per-copy sign/orientation (the edge factor enters inverted) and the Lorentzian edge-mode reading stay [C]. [verification/v138\_vw\_firewall.py]

**Black-hole mechanics in seam units ( $c_3 = 1/(8\pi)$ ): every coefficient is a compiler atom**

| quantity           | standard form                  | seam form                           | compiler atom                                       |
|--------------------|--------------------------------|-------------------------------------|-----------------------------------------------------|
| Einstein equations | $G_{\mu\nu} = 8\pi T_{\mu\nu}$ | $G_{\mu\nu} = T_{\mu\nu}/c_3$       | $c_3 = 1/(8\pi)$ (P1)                               |
| first law          | $dM = \frac{\kappa}{8\pi} dA$  | $dM = c_3 \kappa dA$                | seam constant                                       |
| Smarr (Schw.)      | $M = \frac{\kappa A}{4\pi}$    | $M = 2c_3 \kappa A$                 |                                                     |
| temperature        | $T_H = \frac{\kappa}{2\pi}$    | $T_H = 4c_3 \kappa$                 | $1/(2\pi) = 4c_3$                                   |
| entropy            | $S = \frac{A}{4}$              | $S = 2\pi c_3 A$                    | $1/4 = 1/ \mu_4 $                                   |
| Bekenstein bound   | $S \leq 2\pi ER$               | $S \leq ER/(4c_3)$                  |                                                     |
| Hawking power      | $P = \frac{1}{15360\pi M^2}$   | $P = \frac{c_3}{1920 M^2}$          | $1920 =  W(D_5) $                                   |
| lifetime           | $\tau = 5120\pi M^3$           | $\tau = 128 g_{\text{car}} M^3/c_3$ | $128 = 2^7$ , $7 = \text{scalaron}$                 |
| Kerr extremal      | $A_{\text{ext}} = 8\pi M^2$    | $A_{\text{ext}} = M^2/c_3$          | $A_{\text{ext}}/A_{\text{Schw}} = 1/ \mathbb{Z}_2 $ |
| area quantum (Hod) | $\Delta A = 4\ln 3$            | $\Delta A = \ln(N_{\text{fam}}^4)$  | $81 = \text{disc of the cover}$                     |
| Nariai bound       | $S_N = \frac{2\pi}{\Lambda}$   | $S_N = \frac{2}{3} S_{dS}$          | $2/3 =  \mathbb{Z}_2 /N_{\text{fam}}$ (Koide)       |

Figure 6: Black-hole mechanics in seam units: every classical coefficient is a compiler atom. [verification/v101\_horizon\_anchor.py]

## 6 Turnaround radius, gravitational waves, light

**Bound structures.** In a  $\Lambda$ -universe the maximal bound radius is  $r_{\text{ta}}(M) = (GM[2\pi e^{\alpha_\star^{-1}}/\bar{M}_{\text{Pl}}]^2)^{1/3}$  — halo, cluster, supercluster and void scales follow from the  $\Lambda$  closure with no new parameter. [C]



**Gravitational waves.** The low-curvature branch is  $R + R^2$ ; the tensor graviton runs on the Lorentz cone and the only extra mode (the scalaron,  $M_{\text{scal}} \approx 3.06 \times 10^{13}$  GeV) is frozen for astrophysical frequencies. Hence

$$\boxed{v_{\text{GW}} = c} \quad (\text{no measurable dispersion}) \quad [\text{C}],$$

a clean falsifier: a genuine  $v_{\text{GW}} \neq c$  would break this gravity closure.

**Light.** TFPT does not predict the SI number  $c$  (definitional); it predicts the *universality of the causal cone* — RP  $\rightarrow$  OS reconstruction  $\rightarrow$  one Lorentzian cone shared by EM, gravity and massless modes ( $v_\gamma = v_g = c$ , no native photon dispersion). What it *does* compute is not a speed shift but a polarisation rotation, cosmic birefringence

$$\boxed{\beta_{\text{rad}} = \frac{u}{4\pi} \approx 0.2424^\circ} \quad [\text{E}],$$

from the same seed  $u = \phi_0^{\text{ret}}$  that fixes the Cabibbo angle,  $\Omega_b$  and  $\theta_{13}$ .

## 7 The local sibling: an EHT achromatic polarization intercept

The cosmic birefringence  $\beta_{\text{rad}}$  is the *asymptotic* (Thomson-limit) projection of the determinant-line response; the *local* projection onto the magnetised collar of a horizon-scale black hole is a structured, achromatic residual rotation of the linear-polarization angle [verification/v203\_eht\_achromatic.py]

$$\beta_{\text{BH}}(r) = \frac{Q_e^{\text{eff}}(x) Q_m^{\text{eff}}(x)}{256\pi^4 r^2} = 16 c_3^4 \frac{Q_e^{\text{eff}} Q_m^{\text{eff}}}{r^2} = \frac{\delta_{\text{top}}}{3} \frac{Q_e^{\text{eff}} Q_m^{\text{eff}}}{r^2},$$

where the coupling  $16c_3^4 = 1/(256\pi^4)$  is fixed *exactly* ([E]) — the *same* top-form coefficient  $\delta_{\text{top}} = 48c_3^4 = 3/(256\pi^4)$  that controls the  $\alpha$ -kernel precision-zone correction. There is *no free coupling*: EHT-class polarimetry probes the same admissibility data as the  $\alpha$  row, only projected onto the local collar. What TFPT fixes is the structured  $1/r^2$  shape, the achromaticity, and the sign-flip under effective  $E \cdot B$  reversal, *not* the amplitude  $Q_e^{\text{eff}} Q_m^{\text{eff}}$  (an MHD/GR weight); the channel is therefore typed [C] (shape/sign), not a parameter-free pixel prediction.

**Dated kill test [X].** Build the residual intercept  $\chi_0^{\text{res}} = \chi_0^{\text{obs}} - \chi_0^{\text{GRMHD}}$  after honest GRMHD/synchrotron subtraction; it must pass *three independent nulls* — frequency (achromatic, no  $\lambda^2$  tail), spatial ( $1/r^2$  profile), and sign-flip (reverses under  $E \cdot B$  reversal). A residual statistically consistent with zero across the horizon-scale image, or failing any null, falsifies *this channel* (the compiler core is untouched). *Scope*: the old UFE closed RN-type metric ( $D/r^4$ , regular cores, shadow shift) is *not* adopted — it is superseded by the Nariai/seam=horizon readouts above (v190); only the polarization signature survives.

## 8 Search targets (not claims)

### Audit-level search ansätze [O]

- **Black-hole echoes / horizonless compactness:** any near-horizon echo amplitude ratio would be a natural transfer-rate target,  $\mathcal{A}_{n+1}/\mathcal{A}_n \lesssim (2/3)^6 \approx 0.0878$ . The *same* window  $Q_{\text{geom}} \in [\frac{3}{8}, \frac{1}{2}]$  that the SdS roots trace surfaces independently as the maximum horizonless gravastar compactness  $\mathcal{C} = \frac{3}{8}$  (de Sitter endpoint  $\frac{1}{2}$ ) of Jampolski–Rezzolla (2026); since  $\frac{1}{3} < \frac{3}{8} < \frac{4}{9} < \frac{1}{2}$  such an object is a horizonless light-trapping ECO. **experiments/gravastar-compactness** turns this into an echo template (round-trip delay  $\sim 0.7$  ms at  $62 M_\odot$ , amplitude  $\leq (2/3)^6$ ) — a [C] structural echo, no  $\mathcal{C} \leftrightarrow Q_{\text{geom}}$  map proven. A search ansatz, no prediction.
- **Quantum recovery:** the Page-curve recovery kernel  $I \sim (2/3)^{6n}$  is a falsifiable shape if a boundary-recovery rate is ever measured.
- **FRB repeaters:** fast-radio-burst repeaters are a *preregistered search interface* for the



same boundary-recovery kernel (the frozen ratios  $\{1, (2/3)^6, (1/3)^6\}$  and  $\{1, (2/3)^3, (1/3)^3\}$ ), *not* a horizon claim and *not* part of the verification ledger — the work lives in `experiments/frb-tfpt-signatures` with a frozen hypothesis file and surrogate-calibrated null tests; current public data neither confirm nor refute the kernel (a clean null on echo ratios, a single non-replicated window candidate). Hunting ground, not foundation.

- **Cosmological coupling** ( $w = -1$ ): if the de Sitter–vacuum seam interior (the regular-core reading above) makes a seam black hole carry the  $\Lambda$  vacuum,  $w_{\text{in}} = -1$ , then under Croker–Weiner cosmological coupling  $k = -3w_{\text{in}}$  the mass-growth index is  $k = 3$  exactly, i.e. a population  $w = -1$ . `experiments/ccbh-dark-energy` confronts this with Farrah+2023 ( $k = 3.11 \pm 0.79, -0.14\sigma$ ) as a *contested* downstream bridge and an *alternative reading* of the same  $w = -1$  the DESI  $w$ -watchdog tests — not two independent hits.
- **Cosmic spin parity (frontier)**: the  $\mu_4$  clock and the  $PSL(2, \mathbb{C}) \cong SO^+(3, 1)$  boundary orientation allow at most a tiny global remnant, so TFPT predicts *approximate parity*, not a large handedness excess. `experiments/cosmic-handedness` watches the galaxy-spin asymmetry (Shamir JADES  $\sim 3.3\sigma$  monopole) — most likely a Milky-Way-rotation *dipole* systematic and isotropic in Galaxy Zoo — a frontier watchdog, explicitly *not* promoted.

### Net

All horizon formulas share the one seam normalisation  $1/(2\pi) = 4c_3$ . Black-hole, de Sitter and Unruh temperatures, the Page time, scrambling, the Nariai bound, turnaround radii,  $v_{\text{GW}} = c$  and birefringence are *readouts of one seam constant* — standard physics rewritten in TFPT units, with two genuine compiler fingerprints ( $1920 = |W(D_5)|$  in Hawking power,  $|\mu_4| = 4$  in scrambling). No new gravitational physics is claimed; the value of the reframe is that horizons stop being separate modules.

## Appendix: computational verification

The seam-unit identities of this note (the thermal grammar  $\frac{1}{2\pi} = 4c_3$ , the power denominator  $1920 = |W(D_5)|$ , the de Sitter entropy form, the Page fraction, the scrambling  $|\mu_4| = 4$ , and  $\beta_{\text{rad}} = \varphi_0^{\text{ret}}/4\pi = 0.2424^\circ$ ) are re-derived and machine-checked in `verification/v8_horizon.py`. Run `python verification/v8_horizon.py` (needs `mpmath`); it imports the same  $c_3 = 1/(8\pi)$  primitive as the rest of the suite. The search-target ansätze remain **[O]** and are deliberately not part of the pass/fail checks.